

Git & GitHub Complete Training Guide for Junior Developers

1. Introduction to Git

Git is a distributed version control system used to track changes in source code.
Benefits: collaboration, history tracking, rollback, branching, and code safety.

Version Control Problems Solved:

- Overwriting files
- Losing code
- Team collaboration issues
- Release management difficulties

2. Git Architecture

Working Directory → Staging Area → Local Repository → Remote Repository

Working Directory: Your files.

Staging Area: Prepared changes.

Local Repository: Committed history.

Remote Repository: Shared repository (GitHub).

3. Installation

Windows: Download from git-scm.com

Linux: `sudo apt install git`

Mac: `brew install git`

Verify:

`git --version`

4. Configuration

`git config --global user.name "John Doe"`

`git config --global user.email "john@example.com"`

`git config --list`

5. Repository Creation

`git init`

`git status`

6. Basic Workflow

`git add file.txt`

`git add .`

`git commit -m "Initial commit"`

7. History

```
git log
git log --oneline
git log --graph
git log --all
```

8. File States

Untracked → Modified → Staged → Committed

9. Branching

```
git branch
git branch feature/login
git switch feature/login
git checkout -b feature/login
```

10. Merging

```
git merge feature/login
```

Fast-forward merge and three-way merge concepts.

11. Merge Conflicts

Resolve conflicts manually, then:

```
git add .
git commit
```

12. Undo Operations

```
git restore file.txt
git restore --staged file.txt
git commit --amend
git reset --soft HEAD~1
git reset --mixed HEAD~1
git reset --hard HEAD~1
```

13. Stash

```
git stash
git stash list
git stash apply
git stash pop
```

14. Remote Repositories

```
git remote add origin URL
git remote -v
```

15. Clone

```
git clone URL
```

16. Push

```
git push
git push origin main
git push -u origin main
```

17. Pull and Fetch

```
git pull
git fetch
```

Fetch downloads changes.
Pull downloads and merges.

18. Rebase

```
git rebase main
```

Rewrites history for cleaner commit chains.

19. Cherry Pick

```
git cherry-pick
```

20. Tags

```
git tag v1.0
git tag -a v1.0 -m "Release"
git push origin --tags
```

21. .gitignore

```
node_modules/
bin/
obj/
dist/
.env
```

22. Pull Requests

Create branch → Push → Open PR → Review → Merge

23. Professional Workflow

main
develop
feature/*
release/*
hotfix/*

24. Best Practices

- Small commits
- Meaningful commit messages
- Pull before push
- Never commit secrets
- Use feature branches

25. Interview Questions

1. What is Git?
 2. Difference between Git and GitHub?
 3. Merge vs Rebase?
 4. Fetch vs Pull?
 5. Reset vs Revert?
 6. What is Cherry Pick?
 7. What is Detached HEAD?
 8. What is Stash?
 9. What is Fast Forward Merge?
 10. What is .gitignore?
- (Continue discussion in class with detailed answers.)

26. Git Cheat Sheet

git init
git status
git add .
git commit -m "msg"
git log --oneline
git branch
git checkout -b feature
git merge feature
git stash
git clone URL
git pull
git push
git fetch
git rebase main
git cherry-pick HASH
git tag v1.0
git remote -v